Module 04: Cognition in Robotics

Learning Objectives

1. Define **Cognition** within the context of Robotics.
2. Analyze how Cognition interacts with other components of the Active Inference framework.
3. Apply specific constraints of Robotics to the formal definition of Cognition.

Introduction

This module explores **Cognition**. In the **Robotics** curriculum, we approach this topic with a focus on specific applications and theoretical depth appropriate for the audience. Cognition is a critical component of the 8-part Active Inference spine, bridging the gap between Perception and Action.

Key Concepts

1. Cognition as a Markov Blanket Boundary

How does Cognition define the boundary between the agent and the environment?

2. Generative Models of Cognition

What parameters involved in Cognition must be optimized to minimize variational free energy?

3. Active Inference Dynamics

How does the process of Cognition drive the perception-action loop?

Applications

In Robotics, we see Cognition manifest in: * **Specific Example 1**: A honeybee-inspired navigation robot maintains a cognitive map analogous to the insect's mushroom body, encoding place-action associations learned during exploration as a generative model; when placed in a previously visited environment, the robot recognizes familiar visual scenes (low prediction error) and retrieves associated motor policies, while novel scenes (high prediction error) trigger exploratory behaviors that expand the cognitive map -- mirroring how bees use optic flow and landmark snapshots to navigate between hive and food sources. * **Specific Example 2**: A cephalopod-inspired underwater robot implements distributed cognition where each tentacle-like appendage maintains a local generative model of its contact state and object properties, while a central "brain" module integrates these local beliefs into a global scene understanding; this architecture mirrors the biological octopus's approximately 60% peripheral neural processing and demonstrates how cognition in Active Inference need not be centralized but can emerge from hierarchically composed local inference processes.

Conclusion

Understanding Cognition allows us to better model complex adaptive systems. In the next module, we will expand on this foundation.